

RAG Week Assignment – Section C: Advanced RAG

C1. Multimodal RAG for a Product Manual

Question

You're building a RAG system for a company's product manuals. These manuals have text instructions along with diagrams and photos showing how to assemble the product. A user asks "how do I attach the side panel," and the answer is shown entirely in a diagram, with no accompanying text.

- In multimodal RAG, both text and images are processed and stored for retrieval.
- The user query is converted into an embedding representation.
- The system searches for relevant text as well as related diagrams and images.
- A diagram showing how the side panel is attached can be retrieved even if there is no text explanation.
- The retrieved information is passed to a multimodal model that understands both text and images.
- The model analyzes the diagram and generates an answer for the user.
- In a text-only RAG system, the diagram may never be retrieved because it contains little or no text.
- This can lead to incomplete answers or retrieval failure.

C2. Unified Embedding Space vs Separate Indexes with Fusion

Question

Compare the two approaches and choose one for the product manual scenario.

- In a unified embedding space, text and images are converted into embeddings within the same vector space.
- Related images and text can be retrieved together using a single search.
- In separate indexes with fusion, text and image embeddings are stored in different indexes.
- The system searches both indexes independently and combines the results.
- Unified embedding is simpler because all modalities can be searched together.
- Separate indexes provide more control and flexibility for each data type.
- For the product manual scenario, I would choose unified embedding space.
- A text query such as "attach the side panel" can directly retrieve the related diagram and any matching text from the same retrieval process.